

2012-2013 Fall
MAT123 Final
23/01/2013

1. Determine whether each given sequence with n th term converges or diverges. Evaluate the limit of each convergent sequence. Explain all your work, and write clearly.

a. $a_n = (-1)^n n \sin\left(\frac{1}{n}\right)$

b. $a_n = e^{\cos(\frac{1}{n})}$

2. Determine whether the following series converge or diverge. Give reasons for your answers.

a. $\sum_{n=1}^{\infty} \frac{n}{(3+n^2)^{3/4}}$

b. $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}} \ln\left(\frac{n+1}{n-1}\right)$

c. $\sum_{n=1}^{\infty} \frac{(2n)!}{5^n (n!)^2}$

d. $\sum_{n=1}^{\infty} \frac{1}{n^2} e^{1/n}$

3. Integrate the following functions and write each step in detail.

a. $\int \frac{dx}{e^x + 1}$

b. $\int x \arcsin x \, dx$

4. Find the length of the curve $y = \int_0^x \sqrt{\cos(4t)} \, dt$ for $0 \leq x \leq \pi/8$.

5. For the function $f(x) = \frac{x}{x^2 - 4}$,

- a. Find all the asymptotes of f .
- b. Find the intervals of increase or decrease.
- c. Find the local maximum and minimum values, if any.
- d. Find the intervals of concavity and the inflection points, if any.
- e. Sketch the graph of $f(x)$.

1. a. Evaluate the limit of the non-alternating part at infinity.

$$\lim_{n \rightarrow \infty} n \sin \left(\frac{1}{n} \right) \stackrel{n=\frac{1}{u}}{=} \lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$$

However, for odd values of n , the limit at infinity becomes negative. On the other hand, for even values of n , the limit at infinity becomes positive. Therefore, the limit at infinity does not exist. So, the sequence diverges.

- b. The exponential function e^x and the trigonometric function $\cos x$ are continuous everywhere.

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} e^{\cos \left(\frac{1}{n} \right)} = e^{\cos \left(\lim_{n \rightarrow \infty} \frac{1}{n} \right)} = e^{\cos 0} = e^1 = e$$

The sequence converges to $\boxed{e}$.

2. a. Let $a_n = f(n)$. Define $f(x) = \frac{x}{(3+x^2)^{3/4}}$. The function is continuous for $x > 1$ because the numerator and the denominator are continuous for $x > 1$ and $(3+x^2)^{3/4} \neq 0, \forall x \in \mathbb{R}$.

$$\left. \begin{array}{l} x > 0 \\ (3+x^2)^{3/4} > 0 \end{array} \right\} \text{ for } x > 1 \implies \frac{x}{(3+x^2)^{3/4}} > 0$$

For $x > 1$, $x < x^{3/2} = (x^2)^{3/4} < (3+x^2)^{3/4}$. The denominator grows faster than the numerator. Therefore, the function is decreasing.

We may now apply the Integral Test. Handle the improper integral by taking the limit.

$$\begin{aligned} \int_1^\infty \frac{x}{(3+x^2)^{3/4}} dx &= \lim_{R \rightarrow \infty} 2(3+x^2)^{1/4} \Big|_1^R \\ &= 2 \lim_{R \rightarrow \infty} [(3+R^2)^{1/4} - (3+1^2)^{1/4}] = \infty \end{aligned}$$

Since the integral diverges, the series also $\boxed{\text{diverges}}$.

- b. $\ln(1+x) < x$ for $x > -1$. Therefore,

$$\frac{1}{\sqrt{n}} \ln \left(\frac{n+1}{n-1} \right) = \frac{1}{\sqrt{n}} \ln \left(1 + \frac{2}{n-1} \right) < \frac{1}{\sqrt{n}} \cdot \frac{2}{n-1}$$

Since $n \geq 2$, we have the inequality $2n-2 \geq n \implies \frac{2}{n} \geq \frac{1}{n-1}$.

$$\frac{1}{\sqrt{n}} \cdot \frac{2}{n-1} \leq \frac{1}{\sqrt{n}} \cdot \frac{4}{n} = \frac{4}{n^{3/2}}$$

Now, let $a_n = \frac{1}{\sqrt{n}} \ln \left(\frac{n+1}{n-1} \right)$ and $b_n = \frac{4}{n^{3/2}}$. Apply the Direct Comparison Test.

$$0 < a_n < b_n \implies 0 < \frac{1}{\sqrt{n}} \ln \left(\frac{n+1}{n-1} \right) < \frac{4}{n^{3/2}}$$

b_n converges by the p -series test because $3/2 > 1$. Since b_n converges, by the Direct Comparison Test, a_n also converges.

c. Apply the Ratio Test. Let $a_n = \frac{(2n)!}{5^n (n!)^2}$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(2(n+1))!}{5^{n+1} ((n+1)!)^2} \cdot \frac{5^n (n!)^2}{(2n)!} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(2n+2) \cdot (2n+1) \cdot ((2n)!) \cdot (n!)^2 \cdot 5^n}{5^n \cdot 5 \cdot (n+1)^2 \cdot (n!)^2 \cdot ((2n)!) } \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(2n+2)(2n+1)}{5(n+1)^2} \right| = \lim_{n \rightarrow \infty} \left| \frac{4n^2 + 6n + 2}{5n^2 + 10n + 5} \right| \end{aligned}$$

Now, take the corresponding function and evaluate the limit using L'Hôpital's rule.

$$\lim_{x \rightarrow \infty} \left| \frac{4x^2 + 6x + 2}{5x^2 + 10x + 5} \right| \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \infty} \left| \frac{8x + 6}{10x + 10} \right| \stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \infty} \left| \frac{8}{10} \right| = \frac{4}{5} < 1$$

By the Ratio Test, the series converges absolutely. Since the series converges absolutely, the series converges.

d. Let $a_n = \frac{1}{n^2} e^{1/n}$. Define $f(x) = \frac{1}{x^2} e^{1/x}$. The function is continuous for $x > 1$ because the expressions $\frac{1}{x^2}$ and $e^{1/x}$ are continuous for $x > 1$ and $x^2 \neq 0$, $\forall x > 1$.

$$\left. \begin{array}{l} x^2 > 0 \\ e^{1/x} > 0 \end{array} \right\} \text{ for } x > 1 \implies \frac{e^{1/x}}{x^2} > 0$$

For $x > 1$, $e^{1/x}$ tends to 1 and x^2 is increasing. Therefore, the function is decreasing for $x > 1$.

We may now apply the Integral Test. Handle the improper integral by taking the limit.

$$\int_1^{\infty} \frac{e^{1/x}}{x^2} dx = \lim_{R \rightarrow \infty} -e^{1/x} \Big|_1^{\infty} = \lim_{R \rightarrow \infty} (-e^{1/R} + e^1) = e - 1 \quad (\text{convergent})$$

Since the integral converges, by the Integral Test, the series also converges.

3. a. Add and subtract e^x in the numerator.

$$\int \frac{dx}{e^x + 1} = \int \frac{1 + e^x - e^x}{e^x + 1} dx = \int \frac{e^x + 1}{e^x + 1} dx - \int \frac{e^x}{e^x + 1} dx$$

The result of the integral on the left is x . To calculate the integral on the right, use the u -substitution. Let $u = e^x + 1$, then $du = e^x dx$.

$$x - \int \frac{du}{u} = x - \ln |u| + c = \boxed{x - \ln |e^x + 1| + c = x - \ln (e^x + 1) + c, c \in \mathbb{R}}$$

- b. Solve the integral using the method of integration by parts.

$$\left. \begin{aligned} u = \arcsin x &\implies du = \frac{1}{\sqrt{1-x^2}} dx \\ dv = x dx &\implies v = \frac{x^2}{2} \end{aligned} \right\} \rightarrow \int u dv = uv - \int v du$$

$$\int x \arcsin x dx = \frac{x^2}{2} \arcsin x - \int \frac{x^2}{2\sqrt{1-x^2}} dx \quad (1)$$

Now, we need to find the result of the integral on the right. Let $x = \sin u$ for $\frac{\pi}{2} < u < \frac{\pi}{2}$. Then $dx = \cos u du$. Also, recall the trigonometric identity $\sin^2 x + \cos^2 x = 1$.

$$\begin{aligned} \int \frac{x^2}{2\sqrt{1-x^2}} dx &= \int \frac{\sin^2 u}{2\sqrt{1-\sin^2 u}} \cdot \cos u du = \int \frac{\sin^2 u \cos u}{2\sqrt{\cos^2 u}} du \\ &= \int \frac{\sin^2 u \cos u}{2|\cos u|} du \quad [\cos u > 0] \\ &= \frac{1}{2} \int \sin^2 u du = \frac{1}{2} \int (1 - \cos^2 u) du = \frac{1}{2} \int \left(\frac{1 - \cos 2u}{2} \right) du \\ &= \frac{u}{4} - \frac{\sin 2u}{8} + c = \frac{u}{4} - \frac{\sin u \cos u}{4} + c, \quad c \in \mathbb{R} \end{aligned}$$

Recall the equation $x = \sin u$.

$$x = \sin u \implies \arcsin x = u$$

$$x = \sin u \implies x^2 = \sin^2 u = 1 - \cos^2 u \implies \cos u = \sqrt{1-x^2}$$

Rewrite the result of the last integral.

$$\int \frac{x^2}{2\sqrt{1-x^2}} dx = \frac{1}{4} \left(\arcsin x - x\sqrt{1-x^2} \right)$$

Rewrite (1).

$$\int x \arcsin x \, dx = \boxed{\frac{x^2}{2} \arcsin x - \frac{1}{4} \arcsin x + \frac{1}{4} x \sqrt{1-x^2} + c, \quad c \in \mathbb{R}}$$

4. The length of a curve defined by $y = f(x)$ whose derivative is continuous on the interval $a \leq x \leq b$ can be evaluated using the integral.

$$S = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx.$$

Find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx} \int_0^x \sqrt{\cos(4t)} \, dt$$

By the Fundamental Theorem of Calculus, $\frac{dy}{dx}$ can be rewritten as

$$\frac{dy}{dx} = \sqrt{\cos(4x)}$$

Set $a = 0$, $b = \frac{\pi}{8}$, $\frac{dy}{dx} = \sqrt{\cos(4x)}$ and then find the length.

Recall $\cos(4x) = 2 \cos^2(2x) - 1$.

$$\begin{aligned} S &= \int_0^{\pi/8} \sqrt{1 + \left(\sqrt{\cos(4x)}\right)^2} \, dx = \int_0^{\pi/8} \sqrt{1 + \cos(4x)} \, dx \\ &= \int_0^{\pi/8} \sqrt{2 \cos^2(2x)} \, dx = \sqrt{2} \int_0^{\pi/8} |\cos(2x)| \, dx \quad [\cos(2x) > 0] \\ &= \sqrt{2} \int_0^{\pi/8} \cos(2x) \, dx = \frac{\sqrt{2}}{2} \sin(2x) \Big|_0^{\pi/8} = \frac{\sqrt{2}}{2} \left(\sin \frac{\pi}{4} - \sin 0 \right) = \boxed{\frac{1}{2}} \end{aligned}$$

5. a. Find the horizontal asymptotes.

$$\lim_{x \rightarrow \pm\infty} \frac{1}{x^2 - 4} = 0$$

Find the vertical asymptotes. The expression is undefined for $x = \pm 2$.

$$\lim_{x \rightarrow 2^+} \frac{1}{x^2 - 4} = \lim_{x \rightarrow -2^+} \frac{1}{x^2 - 4} = \infty, \quad \lim_{x \rightarrow 2^-} \frac{1}{x^2 - 4} = \lim_{x \rightarrow -2^-} \frac{1}{x^2 - 4} = -\infty$$

The horizontal asymptote is $y = 0$. The vertical asymptotes are $x = \pm 2$.

- b. Compute the first derivative and set it to 0 to find the critical points. Apply the

product rule appropriately.

$$f'(x) = \frac{1 \cdot (x^2 - 4) - x \cdot (2x)}{(x^2 - 4)^2} = -\frac{x^2 + 4}{(x^2 - 4)^2}$$

f is increasing where $f'(x) > 0$ and decreasing where $f'(x) < 0$. Therefore,

f is decreasing everywhere except at the undefined points.

c. No local maximum or minimum values exist.

d. Compute the second derivative.

$$f''(x) = -\frac{2x \cdot (x^2 - 4)^2 - (x^2 + 4) \cdot 2 \cdot (x^2 - 4) \cdot (2x)}{(x^2 - 4)^4} = \frac{2x^3 + 24x}{(x^2 - 4)^3}$$

An inflection point occurs at $x = 0$.
 f is concave up for $-2 < x < 0 \cup x > 2$.
 f is concave down for $x < -2 \cup 0 < x < 2$.

e. The graph is below.

